5

Citric Acid – Alternative Process using Starch

5.1 Introduction

Citric acid is one of the few commodity chemicals produced in a biotechnical process. The world production is approximately 1.1 million tons per year. Most of the production is used in beverages (45%) and foods (25%) as a flavor enhancer and a preservative. About 20% is used in soaps and detergents. In the chemical and pharmaceutical industry, citric acid is used in buffers, as an antioxidant, flavor additive, and for complexing metals.

A general introduction to citric acid production is given by Kristiansen et al. [5.1]. Citric acid has been produced for over 80 years using the filamentous fungus *Aspergillus niger*. More recently, yeast processes have been used as well. While molasses is a common raw material, in this case study we describe a process that uses pure starch as an alternative carbon source. Data were taken mainly from Marending [5.2]. There are other published citric acid processes starting from starch [5.3–5.5]. The process is described in greater detail by Biwer [5.6] and Biwer and Heinzle [5.7].

5.2 Fermentation Model

Figure 5.1 shows the reaction scheme for citric acid production. In the first step, α-amylase is added to hydrolyse the starch to dextrin. Complete starch hydrolysis cleaves starch into glucose monomers. In the citric acid case, starch is only hydrolysed to dextrin with five glucose units on average. Proteins and fats are common impurities in commercially available starch. We assume that the raw starch used is completely dry and that it contains approximately 1% proteins and 1% fats. The ash content of starch was not considered in this model. These facts have to be taken into account for the definition of the starch price.

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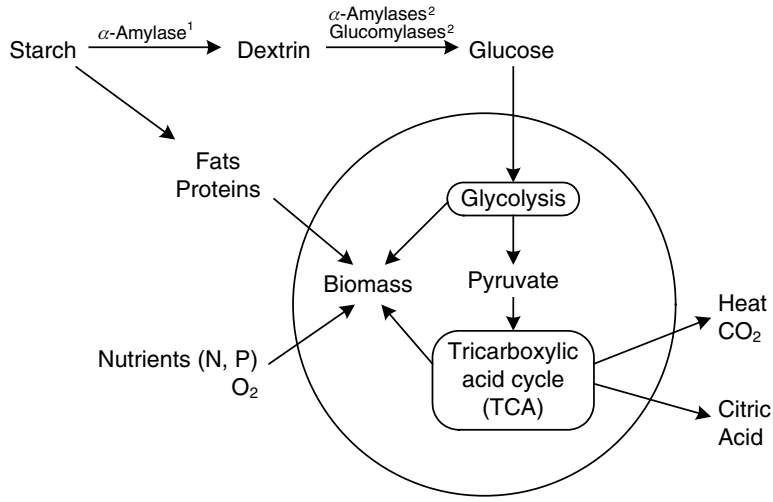


Figure 5.1 *Reaction scheme for citric acid production from starch using* Aspergillus niger. [1] *Added,* [2] *secreted to the medium by* A. niger

The molecular weight of dextrin is 828 g/mol with five glucose units. The degradation of a starch requires one molecule of water for each dextrin molecule formed:

$$(C_6H_{10}O_5)_x + \frac{x}{5} H_2O \rightarrow \frac{x}{5}(C_6H_{10}O_5)_5H_2O \tag{5.1}$$

Hence, 18 g of water (1 mol) are needed for 828 g dextrin, respectively 21.7 g for 1 kg. Thus, 978.3 g pure starch is necessary to obtain 1 kg dextrin. Including the impurities in the starch, the reaction equation is (in g):

$$998.26\,(C_6H_{10}O_5)_x + 21.74\,H_2O \rightarrow 1000.0\,(C_6H_{10}O_5)_5H_2O + 10.0\,\text{Proteins} + 10.0\,\text{Fats} \tag{5.2}$$

Since the exact elementary composition of the proteins and fats contained in raw starch is not known, they cannot be considered in detail for the equation. It is assumed that fats are not modified.

After the starch hydrolysis, temperature and pH are adjusted and the inoculum is added. During the fermentation several reactions run more or less in parallel. The fungus secretes α-amylases and glucoamylases into the media. These enzymes catalyse the degradation of dextrin to glucose that is consumed by *A. niger*. A molar yield of 100% of glucose from starch is assumed.

$$(C_6H_{10}O_5)_5H_2O + 4\,H_2O \rightarrow 5\,C_6H_{12}O_6 \tag{5.3}$$

and in grams:

$$828.7\,(C_6H_{10}O_5)_5 + 72.1\,H_2O \rightarrow 900.8\,C_6H_{12}O_6$$

The glucose is used to form biomass, produce citric acid, and provide energy via the degradation of glucose to carbon dioxide in the respiratory chain. Two phases of cultivation can be distinguished: (i) biomass formation and (ii) citric acid production. However, for the

process modeling, only the final concentration and productivity are relevant and, therefore, only one fermentation step is defined.

Starting concentrations of the raw materials were taken from Marending [5.2]. Ammonium nitrate and protein impurities are used as nitrogen source for the biomass formation. They are both assumed to be consumed completely. The amount of ammonium nitrate added in the medium is known [5.2]. The nitrogen content of the proteins is estimated. Nielsen and Villadsen [5.8] provided the average frequency of the different amino acids in yeast, and Creighton [5.9] calculated the average frequency of amino acids from 1000 known proteins. From the relative frequency and the elementary composition of the amino acids, an average composition was calculated. Thereby, it is assumed that during the polypeptide formation one mol of water is formed per mol of amino acid. The calculated average composition is very similar for the two literature sources. Their average is taken for the following calculations. Referred to one carbon atom, the elementary protein composition is $CH_{1.51}O_{0.3}N_{0.28}$, and the molecular weight is 22.23 g/C-mol. From this composition and the protein amount, the available nitrogen is calculated. In this case 25% of total nitrogen in the biomass is derived from proteins contained in starch and the rest from ammonium nitrate. The sulfur content of the proteins is neglected.

From the available amount of nitrogen and the amount of biomass formed, a nitrogen content of 5.5% is calculated for the biomass. This is lower than the 9.3% typically specified for *A. niger* in literature [5.8]. However, Schlieker [5.10] has shown that the nitrogen content of microbial biomass can substantially decrease under nitrogen limitation. The same is true for the phosphorus content. Here, also the calculated value is relatively small. The estimated elementary composition of the biomass used here is $CH_{1.72}O_{0.55}N_{0.09}P_{0.002}$ (MW = 23.89 g/C-mol).

The reaction equation for the biomass formation from ammonium nitrate is:

$$C_6H_{12}O_6 + 0.28\,NH_4NO_3 + 0.012\,KH_2PO_4 \rightarrow 6\,CH_{1.72}O_{0.55}N_{0.09}P_{0.002} + 1.412\,H_2O + 1.088\,O_2 + 0.012\,K^+ \quad (5.4)$$

The reaction equation for the biomass formation from the proteins is:

$$0.662\,C_6H_{12}O_6 + 2.026\,CH_{1.51}O_{0.3}N_{0.28} + 0.012\,KH_2PO_4 \rightarrow 6\,CH_{1.72}O_{0.55}N_{0.09}P_{0.002} + 0.356\,H_2O + 0.453\,O_2 + 0.012\,K^+ \quad (5.5)$$

For the product formation, glucose is degraded to pyruvate via glycolysis. Pyruvate enters the tricarboxylic acid cycle and is transformed to citric acid that is secreted to the media. The amount of citric acid is expressed as citric acid monohydrate which is the final product.

$$C_6H_{12}O_6 + 1.5\,O_2 \rightarrow C_6H_8O_7 \cdot H_2O + H_2O \quad (5.6)$$

The fermentation ends when the glucose concentration drops below 0.2 g/L. The amount of carbon dioxide produced is estimated via the carbon balance of the fermentation. Glucose start concentration, final citric acid, and start and final biomass concentrations were taken from Marending [5.2], and the overall amounts were calculated for a 210 m^3 working volume (see Table 5.1). 208 334 mol of CO_2 (= 9167 kg) are produced. The reaction equation is:

$$C_6H_{12}O_6 + 6\,O_2 \rightarrow 6\,CO_2 + 6\,H_2O \quad (5.7)$$

Table 5.1 *Estimation of the carbon dioxide produced in the model*

Component	Input (C-mol)	Output (C-mol)
Glucose	1 026 282	1398
Proteins	12 766	
Biomass	352	150 270
Citric acid		679 398
Difference (= CO_2)		208 334

A more detailed description of the citric acid biosynthesis is given in the literature [5.11, 5.12]. A more detailed model would have to consider all impurities as well as moisture content in the raw materials used.

5.3 Process Model

The citric acid production requires a couple of downstream steps following the fermentation. Figure 5.2 shows the process scheme. Based on this process scheme, a process model was developed. For the model, an annual production of 12 000 tons of citric acid is assumed and that is realized with 12 bioreactors, each with a volume of 240 m^3. The number of 12 bioreactors was chosen to facilitate scheduling optimization with a minimal idle time of the downstream equipment.

The corresponding process flow diagram is shown in Figure 5.3. The key process step is the bioreactor (P-6). Starch, water, and amylase (S-109 to S-111) are first added to the reactor, where starch is hydrolysed. Then the bioreactor is filled with medium (from tank P-1) and water (S-107). Both streams are sterilized in continuous heat sterilizers

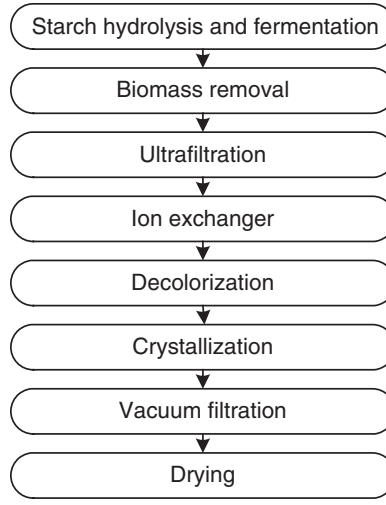


Figure 5.2 *Process scheme of the citric acid production (data taken from [5.2])*

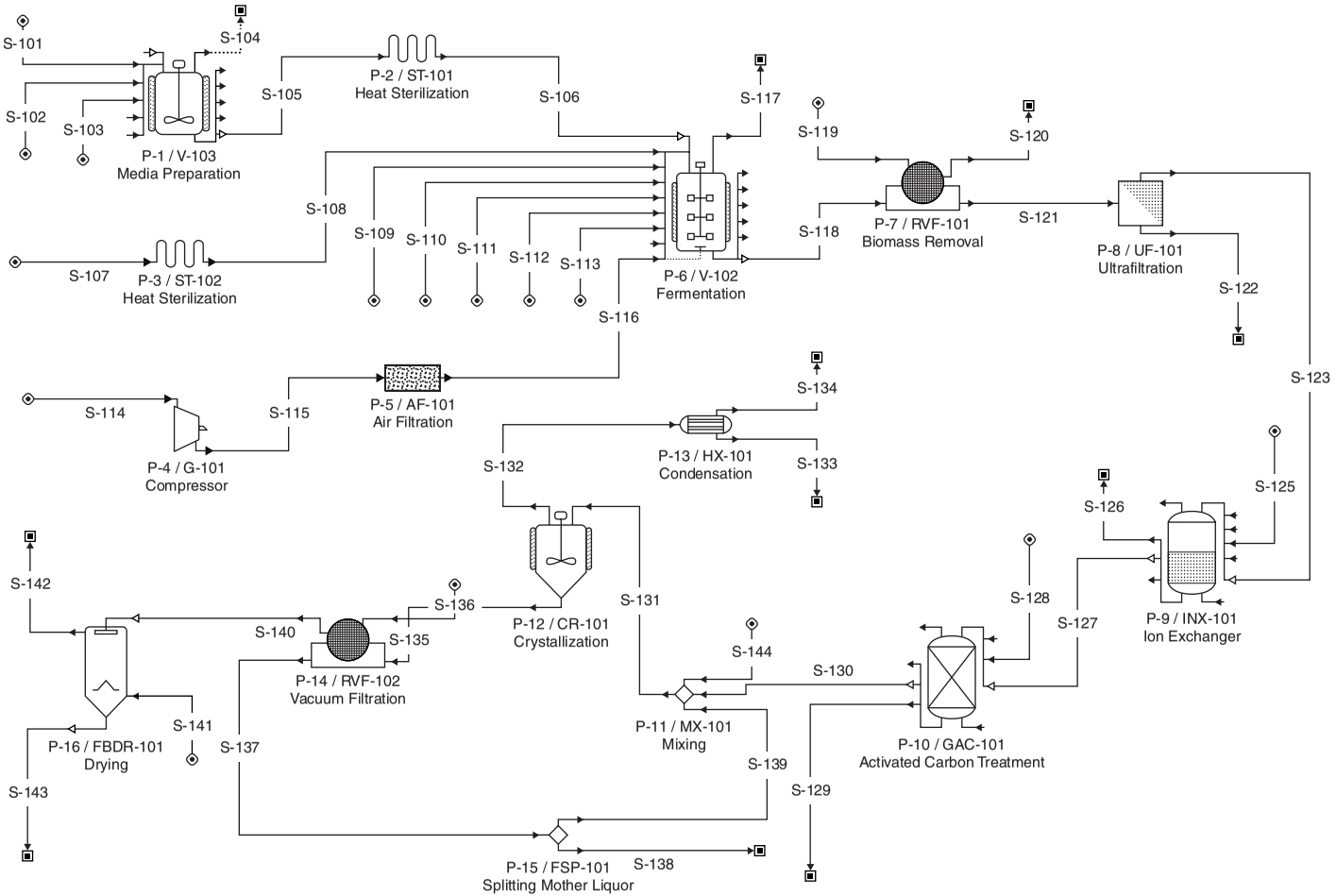


Figure 5.3 *Process flow diagram of the citric acid process*

(P-2, P-3). The pH is adjusted using hydrochloric acid (S-112) and the inoculum is added (S-113). The raw materials for starch hydrolysis do not need to be led through the sterilizer, because the temperature profile of the starch hydrolysis already meets sterilization requirements. During fermentation, the bioreactor is aerated (S-116). Air is supplied by the compressor P-4 and sterilized by the filter P-5.

After the end of the fermentation, proteases secreted to the media by the fungus are inactivated by heat. The bioreactor contents are led to the rotary vacuum filtration P-7 where most of the biomass is removed. The separated biomass is washed to reduce product loss (S-119). Remaining cells, cell debris, and proteins are retained in a subsequent ultrafiltration step (P-8). In the next step, magnesium and potassium ions are separated from the product stream in the ion-exchange column P-9. Cations are bound to the resin, then eluted using hydrochloric acid (S-125) and discharged (S-126), while the product and anions flow through the column. It is assumed that the anions do not affect the crystallization. The product solution is decolorized with activated carbon packed in column P-10. In P-11, caustic soda is added to prevent the evaporation of hydrogen chloride during crystallization.

In the crystallization tank P-12 most of the water is evaporated. Then the solution is cooled and citric acid crystallizes (data from [5.13]). The evaporated water is condensed in P-13. The citric acid crystals are separated and washed (S-136) in the rotary vacuum filter P-14. The mother liquor (S-137) is recycled to the crystallization tank to increase the recovery yield. A part of the mother liquor (S-138) is purged in P-15 to avoid the accumulation of undesired substances. Following Marending [5.2], a crystallization yield of 98% is assumed, although data from Gluszcz and Ledakowicz [5.14] indicate that the yield might be lower. Since citric acid has a high solubility (59% w/v), around 9 kg water/kg citric acid have to be evaporated to realize a high yield at the given product concentration of the feed. Glucose, fats, sodium, and chloride are the main impurities in the feed to the crystallizer. They all remain well below their maximum solubility, and it is assumed that they are separated with the bleed stream S-138.

The recovered crystals as citric acid monohydrate are dried in the fluid bed dryer P-16 using preheated air (S-141). For a 240 m^3 bioreactor, 28.3 tons/batch of starch are consumed and 22.4 tons/batch of citric acid monohydrate (= 20.4 tons pure citric acid) are obtained in the final product (S-143).

5.4 Inventory Analysis

One batch takes 189 h, with the bioreactor occupying 164 h and the downstream processing 34 h. A new batch is started every 14 h in one of the 12 bioreactors. The bioreactors are the bottleneck of the process. With 330 operating days, 12 630 tons of citric acid monohydrate are produced in 565 batches. This assumes that all batches are successful and meet the average target value.

From 100 kg starch, 79 kg of citric acid monohydrate are produced. The carbon yield of the process is 61% (C-mol citric acid/C-mol glucose). The respective yields of the bioreaction are 84% (kg/kg) and 65% (C-mol). The downstream processing yield is 94% with a product loss of 2% in the crystallization and 1% in the biomass removal, the ultrafiltration, and the two adsorption steps each.

Table 5.2 *Material balance of citric acid production. (kg/kg P) = kg component per kg final product (citric acid monohydrate)*

Component	Input (kg/kg P)	Output (kg/kg P)
α-Amylase	<0.01	<0.01
Ammonium nitrate	0.02	
Biomass	<0.01	0.16
Carbon dioxide		0.41
Chloride		<0.01
Citric acid monohydrate (product)		1.00
Citric acid (loss)		0.07
Fats		0.01
Glucose		<0.01
Hydrogen chloride	<0.01	
Sodium hydroxide	<0.01	
Sodium (dissolved)		<0.01
KH_2PO_4	<0.01	
Magnesium sulfate	<0.01	
Magnesium (dissolved)		<0.01
Oxygen	0.51	
Potassium (dissolved)		<0.01
Starch	1.27	
Sulfate		<0.01
Water	14.98	15.12
Mass Index (including water)	16.8	16.8
Mass Index (without water)	1.8	1.65

Table 5.2 shows the overall material balance for the process. The major input materials are typical for a bioprocess: A large amount of water, starch as the carbon source, oxygen for the respiration of the fungus, and a source of nitrogen to support biomass growth. All other compounds are needed in only small amounts. The overall material intensity is 16.8 kg/kg final product (see Table 5.2). Besides water and the product, biomass, carbon dioxide, and the product loss dominate the output. Additionally, several inorganic salts leave the process in smaller amounts.

The majority of the waste is accumulated as wastewater containing relatively low concentrations of organic materials and inorganic salts and must be led to a sewage-treatment plant. The chemical oxygen demand (COD) is around 65 g O_2/kg final product (kg P). Biomass is separated as solid waste in the first vacuum filtration (COD: 280 g O_2/kg P). It can be further used (e.g. as animal feed), added to a wastewater-treatment plant as carbon source, or else is disposed of. Gaseous emissions leave the process from the bioreactor (containing carbon dioxide) and from the dryer and the condenser (water-saturated air). They do not need further treatment. None of the waste streams contains critical materials in relevant amounts. We are assuming no need for odor control.

The process requires substantial amounts of steam, electricity, and cooling and chilled water. Around 10 MJ of electricity is needed per kg final product (2.65 kWh/kg P) with the main consumptions for the air compressor and the bioreactor agitation. The steam

requirement is 32 MJ/kg P (9 kWh/kg P), respectively 15 kg/kg P. Most of it is consumed for water evaporation in the crystallization step and also for the different heating operations in the bioreactor and the dryer. Cooling and chilled water are needed to condense the water vapor in P-13 and also to cool the bioreactors and the compressors, altogether 1.7 m^3/kg P. Air compression (P-4) and evaporation (P-12) are the dominating energy-consumption steps. The aeration rate and the fermentation duration determine the demands on the compressor. The amount of water that has to be evaporated in the crystallization step arises from the citric acid concentration in the feed after the adsorption steps (P-9, P-10).

5.5 Environmental Assessment

The environmental indices are summarized in Table 5.3. Four components are categorized at least once as class A (high environmental relevance). Concentrated hydrogen chloride and sodium hydroxide have a high acute toxicity, while the ammonium nitrate used as nitrogen source is classified A in the category Thermal Risk, because it can be explosive when mixed with flammable substances (R-code 9). A careful handling of these three substances in the process can minimize the risk. In the output, phosphate is classified A due to its importance for eutrophication processes. However, it leaves the process only in very small amounts.

When using the EF_{Mult}, the weighting factor for class C is one. This means the minimal possible $EI_{\text{Mult}} = MI$. When calculating EF_{MV}, class C is set to zero. Here, the minimal possible EI_{MV} is zero. The environmental indices, both for input and output, are all quite close to their minimum possible values (see Table 5.3); this indicates a generally low environmental relevance of the substances involved in the process.

The relative importance of the different components is shown in Figure 5.4 by the EI_{MV} and the EI_{Mult} of the process. Starch, oxygen, and ammonium nitrate are the three dominating components in the input EI_{Mult}. The EI_{MV} does not consider substances with a very low environmental relevance even if they are consumed in high amount. Therefore, starch and oxygen ($EF_{\text{MV}} = 0$) are not included in the input EI_{MV} and ammonium nitrate

Table 5.3 *Environmental assessment parameters of the citric acid process*

	Input		Output	
Assessment parameter	Including water	Without water	Including water	Without water
Mass Index *MI* (kg/kg P)	16.8	1.8	16.8	1.7
Number of A-components	3		1	
Environmental Index EI_{MW} (Index Points/kg P)	0.01		0.05	
Environmental Index EI_{Mult} (Index Points/kg P)	16.9	2.0	17.0	1.9
General Effect Index GEI_{MW} (0–1)	0.0006	0.005	0.003	0.032
General Effect Index GEI_{Mult}	1.01	1.09	1.01	1.13

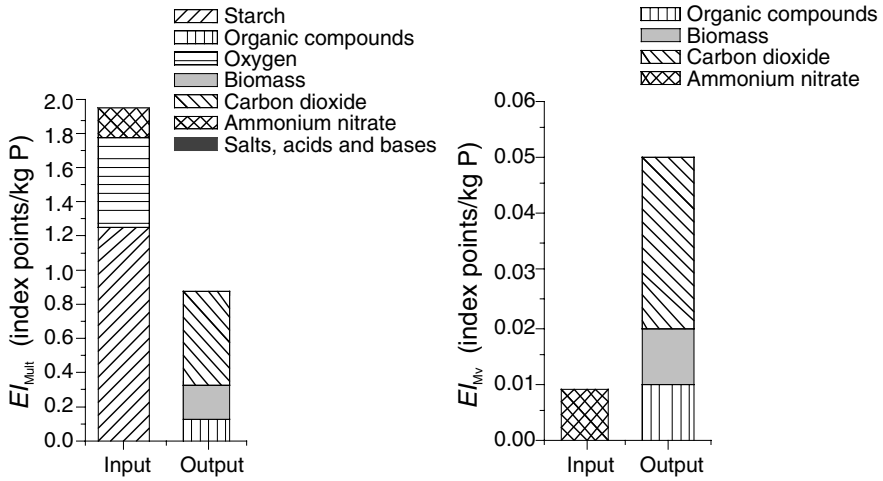


Figure 5.4 *Environmental Indices (EI_{Mult}, EI_{MV}) of the citric acid production. The final product is not considered in the graph*

is the only relevant component besides small amounts of acids and bases (HCl, NaOH). Carbon dioxide, biomass, and organic compounds (product loss, glucose, fats, etc.) are the dominating output components for both indices.

The impact group indices (*IGI*) show how strong the different impact groups contribute to the overall environmental impact of the process. The IG Risk dominates the *IGI* of the input components (see Figure 5.5), while the remaining three IGs (Organisms, Resources, Grey Inputs) are evenly weighted. The dominance of the risk IG is caused by ammonium nitrate (explosive, see above). The IG Risk does not contribute to the environmental relevance of the output components and the IG Organisms only to a small extent (no risk-relevant or toxic substances in the output). The impact groups Air and Water/Soil are relevant. The global warming potential of the carbon dioxide accounts for the importance of the IG Air,

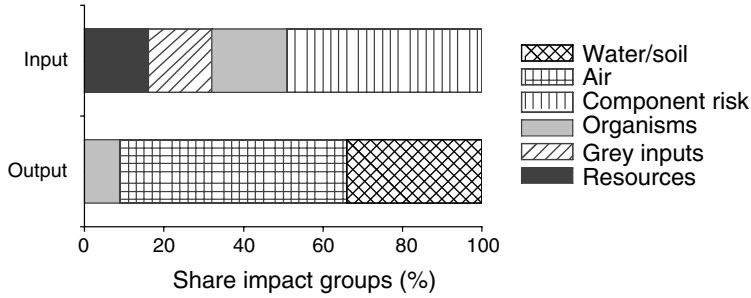


Figure 5.5 *Impact group indices of the input and output components of the citric acid production*

while the eutrophication and the organic carbon pollution potential of the biomass and the organic compounds are responsible for the importance of the IG Water/Soil.

5.6 Economic Assessment

With the model parameters chosen, the plant for the production 12 600 metric tons requires a direct fixed capital investment of $115 million and a total capital investment of $121 million. The most expensive pieces of equipment are the 12 bioreactors and the associated compressors.

The annual operating costs are $31 million. Figure 5.6 shows the allocation to the main cost categories. The facility-dependent costs are the dominating cost type, with the depreciation cost accounting for the largest part. Utilities, raw materials, and labor costs also contribute substantially, while the share of the waste treatment, consumables, and laboratory/QC/QA is small. The fermentation section accounts for 78% of the operating cost, mainly because of the high contribution to the facility-dependent cost (investment for bioreactors and compressors) and the raw material costs (starch).

Starch makes up almost 90% of the raw material costs. Additionally, the amylase, ammonium nitrate, and process water have a notable share. The labor costs are dominated by the fermentation section, almost exclusively caused by the labor demand to run the bioreactors. The waste-treatment costs are evenly divided between the solid waste and the wastewater.

In the model, the overall unit production costs are $ 2.5/kg, which is above the current citric acid selling price of around $ 1.8/kg (2005). Harrison et al. [5.15] found a slightly lower UPC for a citric acid process using molasses and year 2000 prices ($ 2.2/kg). Most citric acid plants operating today are already fully depreciated. Since the depreciation cost accounts for a substantial part of the operating cost, these plants can produce at lower costs, although their maintenance costs are surely higher than in a new plant. Furthermore, the model presented is a generalized representation of a citric acid process. Depending on the specific situation and location of a producer, the costs might be different. Access to inexpensive raw materials and inexpensive equipment is crucial. There is likely to be a strong competitive position for a manufacturer in an advantaged position with regard to raw materials.

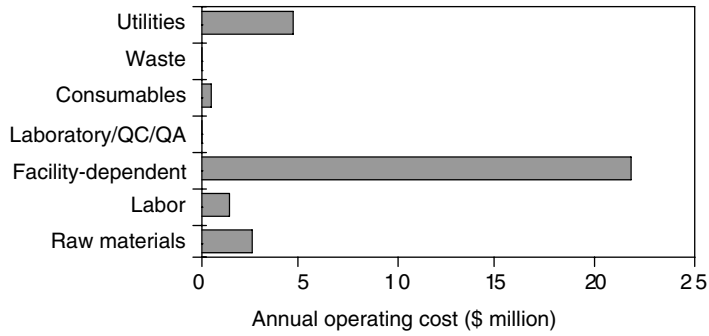


Figure 5.6 *Allocation of the annual operating costs of the citric acid production*

5.7 Conclusions

We modeled the production of citric acid with the use of starch instead of molasses. The use of starch necessitates a different downstream processing for the purification of the citric acid.

The process shows a relatively low potential environmental burden; the ammonium nitrate in the input, and the carbon dioxide emission, the biomass, and the organic compounds in the output, are the most relevant components. The energy consumption also substantially contributes to the environmental burden and to the operating costs. The key factors for a lower environmental impact and lower raw material and utility costs are a higher product concentration, a higher yield, and a shorter fermentation time; this is typical for a commodity process. For a new plant, low investment cost is crucial, e.g. through cheaper equipment acquired from an already depreciated plant or possibly by the replacement of stirred tank bioreactors with alternative reactor types, e.g. bubble columns.

Suggested Exercises

1. Study the influence of final product concentration. What is the influence of an increase in the final product concentration by 10%? Do this by increasing the feed starch concentration (S-109). First watch how biomass concentration, carbon dioxide production, and citric acid yield vary in the bioreactor (S-118). What is the impact on the size of the downstream equipment, unit production costs, and the environmental indices?
2. Assume the strain-development group has identified a new strain with higher productivity reaching the same product yield and concentration in a shorter time. The resulting process time for the fermentation (P-6) decreases from 145 h to 130 h. Examine the same parameters as in Exercise 1 and compare the results.

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